Piatetski–Shapiro primes
Compact Lean blueprint

Manuscript of Hamed Mousavi; project checkpoint, 2 November 2026.

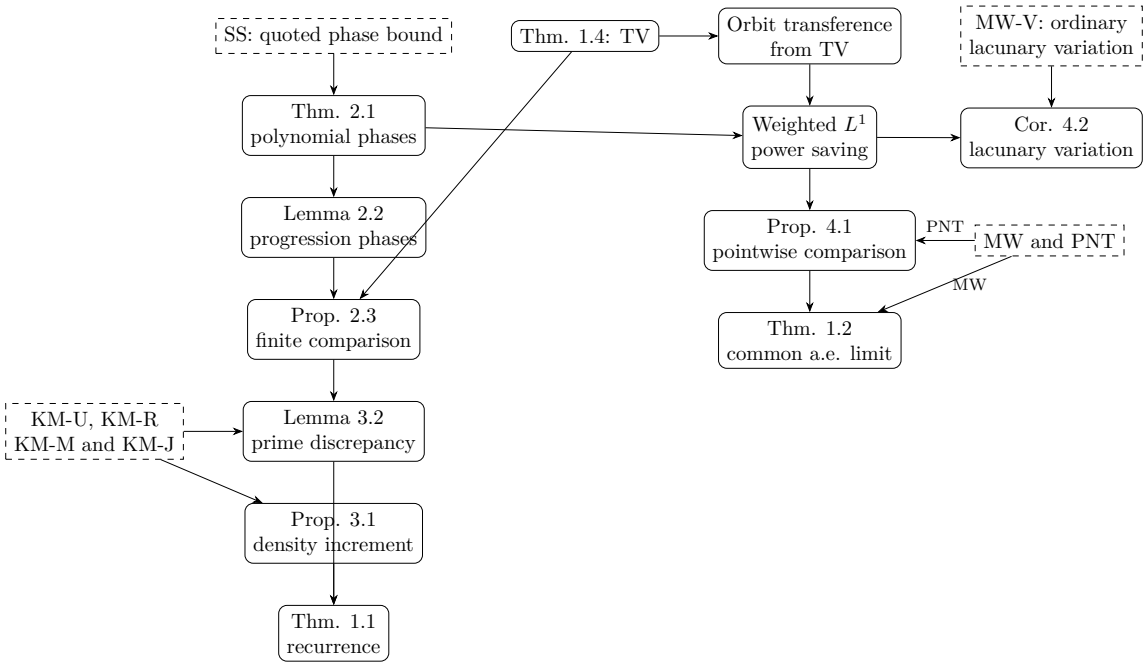
Reading and verification contract

This is the primary reading guide for Hamed Mousavi's *Polynomial configurations and pointwise averages along Piatetski–Shapiro primes*. Current source: unchanged revision supplied 2 November 2026, `manuscript/revised-2026-11-02/Draft.tex`. The original 30 October source and earlier blueprints remain preserved separately. No independent supplied PDF was available. The source is preserved unchanged; its locally compiled PDF supplies the numbering.

Verified boundary. The seven original numbered deductions are checked conditionally. Theorems 1.4 and 2.1 remain quoted external inputs. The external predicates are uninstantiated, and their source-formulation bridges are stated explicitly. Neither an axiom audit nor this blueprint proves those premises true. The revision adds Corollary 4.2 and the separate MWW variation input MW-V; its checked conditional proof is described below.

The related paper's required orbit argument is now proved from TV; its main results are not inputs. Four milestones organize the dependencies: exact objects and algebra; finite comparison; density increment and recurrence; pointwise comparison and unweighting. The longer companion explains the substantive steps and the remaining external obligations.

Dependency graph



Dashed boxes group external inputs. Solid boxes name mathematical steps. Status text distinguishes quoted results and checked conditional deductions. The variation branch uses the L^1 saving directly and does not depend on PNT or the pointwise MW input.

1 Main statements and common objects

Definition D.1 (The polynomial family and the two prime weights)**.** Fix $\ell \geq 1$ and $P_0 = 0$, $P_i \in \mathbb{Z}[y]$, with $P_i(0) = 0$ and

$$1 \leq \deg P_1 < \cdots < \deg P_\ell = d.$$

For $c > 1$, $\gamma = 1/c$, put

$$\mathcal{S}_c = \{\lfloor n^c \rfloor : n \in \mathbb{N}_{>0}\}, \quad \mathbb{P}_c = \mathcal{S}_c \cap \mathbb{P}, \quad \Lambda_\gamma(n) = \gamma^{-1} n^{1-\gamma} \Lambda(n) \mathbf{1}_{\mathcal{S}_c}(n), \quad D_\gamma = \Lambda_\gamma - \Lambda.$$

Weights are zero at nonpositive integer arguments. Set

$$\|w\|_{u^s[M]} = \sup_{\substack{\phi \in \mathbb{R}[y] \\ \deg \phi < s}} \left| M^{-1} \sum_{m=1}^{M} w(m) \mathrm{e}(-\phi(m)) \right|, \qquad \mathrm{e}(t) = \exp(2\pi i t).$$

Status: prime and polynomial objects implemented; the displayed u^s supremum is manuscript notation only.

Theorem 1.1 (Quantitative recurrence; manuscript Theorem 1.1)**.** *For every fixed family above there exist $c_* > 1$ and $\kappa > 0$ such that, for every $1 < c < c_*$, there is $N_0(\mathbf{P}, c)$ with the following property. If $N \geq N_0$ and $A \subset [N]$ satisfies*

$$|A| \geq N/(\log N)^\kappa,$$

then, for some $p \in \mathbb{P}_c$ and $x \in \mathbb{Z}$, all the elements

$$x,\ x + P_1(p-1), \ldots, x + P_\ell(p-1)$$

belong to A and are pairwise distinct.

Status: checked conditional proof from SS, TV, KM-R and KM-Compatible.

Theorem 1.2 (Pointwise prime averages; manuscript Theorem 1.2)**.** *After decreasing $c_* > 1$ if needed, for every $1 < c < c_*$, every invertible probability measure-preserving system (X, ν, T) and every $f_1, \ldots, f_\ell \in L^\infty(X)$, the averages*

$$\frac{1}{\pi_c(N)} \sum_{\substack{p \leq N \\ p \in \mathbb{P}_c}} \prod_{i=1}^{\ell} f_i(T^{P_i(p-1)} x)$$

converge a.e. to the same limit as the analogous average over all primes, normalized by $\pi(N)$. Here $\pi_c(N) = |\mathbb{P}_c \cap [N]|$.

Status: checked conditional proof from SS, TV, PNT and MW; full uniform c_* specification.

Theorem 1.3 (Finite polynomial estimate; manuscript Theorem 1.4)**.** *For fixed integer polynomials $R_1, \ldots, R_\ell$ of distinct increasing degrees, largest degree d, and $C_0 \geq 1$, there are $K \geq 1$, $K \ll_d 1$, and a constant $C = C(C_0, \mathbf{R})$ such that, for all $M \geq 1$, 1-bounded F_i supported on $[-C_0 M^d, C_0 M^d]$ and 1-bounded θ on $[M]$,*

$$\left| M^{-d-1} \sum_{x \in \mathbb{Z}} \sum_{m=1}^{M} \theta(m) F_0(x) \prod_{i=1}^{\ell} F_i(x + R_i(m)) \right| \leq C\big(M^{-1} + \|\theta\|_{u^{d+1}[M]}\big)^{1/K}.$$

Status: external input TV, not a proved project theorem. The actual `FiniteVNInput` uses universal phase bounds and integer scales, with positive degree. Its source-to-interface bridge TV-B (supremum, sign and monotonicity) is explicitly unformalized. Its constant depends on the fixed family. Scaling $\theta = g/B$ gives the bound $CB(M^{-1} + B^{-1}\|g\|_u)^{1/K}$ when $|g| \leq B$ and $B > 0$. This scaling is an elementary deduction, not a strengthened external input.

2 Arithmetic and finite comparison

Theorem 2.1 (Polynomial-phase saving; manuscript Theorem 2.1)**.** *For every integer $r \geq 1$ there are $\varepsilon_r, \sigma_r > 0$ such that, whenever $1 - \varepsilon_r < \gamma < 1$,*

$$\sup_{\deg \phi \leq r} \left| \sum_{n=1}^{X} D_\gamma(n) \mathrm{e}(\phi(n)) \right| \ll_{r,\gamma} X^{1-\sigma_r}.$$

The implied constant is uniform over all real polynomial coefficients.

Status: external input SS: directly quoted polynomial-phase consequence.

Mathematical derivation and formalization boundary. Take $G = \mathbb{R}$, $\Gamma = \mathbb{Z}$, $G_0 = \cdots = G_r = \mathbb{R}$ and $G_j = \{0\}$ for $j > r$. Set $F(t + \mathbb{Z}) = \mathrm{e}(t)$. The filtered nilmanifold and Lipschitz bounds depend only on r. Expand $\phi(n) = \sum_{j=0}^{r} a_j \binom{n}{j}$. This is an adapted polynomial sequence, so SS gives the estimate with constants independent of the a_j. No quantitative Gowers-uniformity statement for D_γ is used. □

Lemma 2.2 (Restriction to progressions; manuscript Lemma 2.2)**.** *With the same admissible γ range and some positive saving σ_r, for all integers $Q, M \geq 1$,*

$$\sup_{\deg \psi \leq r} \left| \sum_{y=1}^{M} D_\gamma(Qy + 1) \mathrm{e}(\psi(y)) \right| \ll_{r,\gamma} (QM)^{1-\sigma_r}.$$

Status: checked conditional proof from `ShaoSunPhaseInput`; saving may be decreased.

Restriction argument. Write $L = QM$ and reindex by $n = Qy + 1$. Insert

$$\mathbf{1}_{n \equiv 1 \pmod{Q}} = Q^{-1} \sum_{a=0}^{Q-1} \mathrm{e}(a(n-1)/Q).$$

For each a the phase $\psi((n-1)/Q) + a(n-1)/Q$ has degree at most r, because $r \geq 1$. Including $n = 1$ causes no error since $D_\gamma(1) = 0$. Apply Theorem 2.1 at $L + 1$ to each character and average; the factor Q cancels exactly. □

Definition D.2 (Rescaling and the finite counting form)**.** For $P(0) = 0$ and integer $Q \geq 1$, define the integer polynomial

$$P_Q(y) = P(Qy)/Q = \sum_{j=1}^{\deg P} a_j Q^{j-1} y^j, \qquad P(y) = \sum_{j \geq 1} a_j y^j.$$

For finitely supported $f_i : \mathbb{Z} \to \mathbb{C}$ set

$$\mathcal{T}_{N,M,Q}(w; \mathbf{f}) = \frac{1}{NM} \sum_{x} \sum_{y=1}^{M} w(Qy + 1) f_0(x) \prod_{i=1}^{\ell} f_i(x + P_{i,Q}(y)).$$

Status: exact algebra checked; parent Proposition 2.3 conditionally checked.

Proposition 2.3 (Finite comparison; manuscript Proposition 2.3)**.** *Fix $C_0 \geq 1$, $0 < a < 1$ and the original family. There are $c_*(d) > 1$ and $\xi(d) > 0$ such that, for $1 < c < c_*$,*

$$Q \leq \exp((\log N)^a), \qquad M = \left\lfloor (N/Q^{d-1})^{1/d} \right\rfloor$$

imply, for sufficiently large M in terms of $c, \mathbf{P}, C_0, a$ and 1-bounded inputs supported on $[-C_0 N, C_0 N]$,

$$|\mathcal{T}_{N,M,Q}(D_\gamma; \mathbf{f})| \ll_{c,\mathbf{P},C_0} N^{-\xi}.$$

Status: checked conditional proof from SS and TV; TV-B remains external.

Fixed-family lift. Set $L = QM$. Lift f_i by $F_i(Qx) = f_i(x)$ and $F_i(z) = 0$ otherwise. Set

$$B = C_\gamma L^{1-\gamma}\log(2L), \qquad \Theta(n) = B^{-1}D_\gamma(n+1)\mathbf{1}_{Q|n}.$$

Then $|\Theta| \le 1$ and

$$\|\Theta\|_{u^{d+1}[L]} \ll B^{-1}L^{-\sigma_d}.$$

The lifted count uses the original fixed P_i and has normalization loss

$$\frac{L^{d+1}}{NM} \asymp Q^2.$$

TV therefore bounds the original count by

$$\ll BQ^2\big(L^{-1} + B^{-1}L^{-\sigma_d}\big)^{1/K}.$$

Choose $1-\gamma < \min(1,\sigma_d)/(8K)$; use $B \ll_\gamma L^{2(1-\gamma)}$ and absorb $Q^2 = L^{o(1)}$. The remaining negative power of $L = N^{1/d+o(1)}$ gives the claimed saving. □

3 Density increment and recurrence

Definition D.3 (Coefficient and modulus conditions)**.** Write $\mathcal{L}(s) = \operatorname{lcm}(1,\dots,s)$ and $P^+(1) = 1$. A positive-degree polynomial $R(y) = \sum_{j=1}^{e} a_j y^j$ has (C,Q)-coefficients if

$$a_e = a'_e Q^{e-1}, \quad 0 < |a'_e| \le C, \quad |a_j| \le C|a_e| \quad (1 \le j < e).$$

A single C bounding all coefficients of the original family works for every rescaling $P_{i,Q}$ and every positive Q.

Proposition 3.1 (Discrepancy gives an increment; manuscript Proposition 3.1)**.** *For fixed C, d, ℓ there are $C_1, K_1 \ge 2$, $0 < c_0 < 1/2$ and $c_1, c_2 > 0$ such that the following holds for sufficiently large N. Suppose*

$$(\log N)^{-1/C_1^2} < \delta \le 1, \quad \lceil K_1\delta^{-C_1}\rceil \le s \le (\log N)^{c_0},$$
$$Q \le \exp((\log N)^{2c_0}), \quad P^+(Q) \le s, \quad \mathcal{L}(s) \mid Q.$$

Let R_i have the specified distinct degrees and (C,Q)-coefficients, $R_0 = 0$, and put $M = \lfloor (N/Q^{d-1})^{1/d}\rfloor$. If $|A| = \delta N$ and the absolute prime-weighted discrepancy between $\mathbf{1}_A$ and $\delta\mathbf{1}_{[N]}$ is at least

$$c_1\delta^{\ell+1}\mathbb{E}_{y\le M}\Lambda(Qy+1),$$

then an atom $a + Q^b[N'] \subset [N]$, with $1 \le b \ll_d 1$, has density at least $(1+c_2)\delta$ and length

$$N' \ge Q^{-O_{d,\ell}(1)}\delta^{O_{d,\ell}(1)}M^r\exp(-d\sqrt{\log M}), \qquad r = \deg R_1.$$

Status: checked conditional proof from KM-U and KM-R, including literal length normalization.

The original local-factor deduction. KM-U gives $Me^{-\sqrt{\log M}} \le M_0 \le M$ and unweighted discrepancy at least $(c_1/4)\delta^{\ell+1}$. Apply KM-R with $\varepsilon = (c_1/16)\delta^{3(\ell+1)}$. Its small-scale alternative contradicts $\log M_0 \gg_d \log N$. Otherwise let F_i be the densities on its progression partitions. They preserve mass, and the approximation leaves discrepancy at least $(c_1/8)\delta^{\ell+1}$.

If every $F_i \le (1+c_2)\delta$, equality of masses gives

$$\sum_x |F_i(x) - \delta \mathbf{1}_{[N]}(x)| \le 2c_2\delta N.$$

Telescoping the products bounds the discrepancy by $2(\ell+1)c_2(1+c_2)^\ell\delta^{\ell+1}$, a contradiction when c_2 is small. Some atom therefore has the required density. KM-R's resolution and the bound on M_0 give the length estimate. □

Lemma 3.2 (Sparse-prime absence forces discrepancy; manuscript Lemma 3.2)**.** *Fix the original family and the constants of Proposition 3.1. There is a degree-dependent $c_* > 1$ such that, for $1 < c < c_*$ and sufficiently large N, the parameter hypotheses of Proposition 3.1 imply the following. If $A \subset [N]$ contains no pairwise-distinct configuration*

$$x,\ x + P_{1,Q}(y), \dots, x + P_{\ell,Q}(y), \quad y \ge 1, \quad Qy + 1 \in \mathbb{P}_c,$$

then its prime-weighted discrepancy satisfies the hypothesis of Proposition 3.1.

Status: checked conditional proof from SS, TV and compatible same-parameter prime-mass data.

Main term and the three errors. Choose small fixed $\tau > 0$. For $y \le \lfloor \tau M \rfloor$, all shifts have absolute value at most $N/10$, so at least $N/2$ base points survive. The ordinary-prime model count is therefore $\gg_{\mathbf{P}} \delta^{\ell+1}\mathbb{E}_{y\le M}\Lambda(Qy+1)$, by the prime-mass truncation comparison.

Comparison changes the count in A to the Λ_γ count with error $O(N^{-\xi})$. Its noncollision prime terms vanish by hypothesis. Collisions occur at $O_{d,\ell}(1)$ roots of nonzero difference polynomials, contributing

$$\ll (QM)^{1-\gamma}\log(2QM)/M.$$

Higher prime powers contribute

$$\ll Q^{3/2-\gamma}M^{1/2-\gamma}(\log(2QM))^2.$$

All three errors are $O(N^{-\eta})$ for a positive η. The lower prime-mass bound $\mathbb{E}\Lambda(Qy+1) \gg \exp(-O((\log N)^{2c_0}))$ and the density range make the errors negligible compared with the model. Subtracting the actual count from that model proves the discrepancy. □

Iteration proving Theorem 1.1

Fix $s = \lceil K_1\alpha^{-C_1} \rceil$ and $Q_0 = \mathcal{L}(s)$, and start on a densest full residue-class intersection. At every increment set $Q_{j+1} = Q_j^{b_j+1}$ and pull A_j back to the selected progression. Exact nested rescaling preserves absence and distinctness. The modulus remains a power of Q_0; s remains fixed. Quantitative length recursion and $\log Q_j \le \alpha^{-O(1)}$ show that every step is admissible when $\alpha \ge (\log N)^{-\kappa}$ and κ is small enough. After $O(\log\alpha^{-1}+1)$ increments, the density exceeds 1. The actual finite-prefix construction and uniform admissibility proof are checked; the increment provider is discharged using the quoted source estimates and the original discrepancy argument.

Checked assembly of Theorem 1.1. `quantitative_recurrence` chooses the maximum degree and a coefficient bound, then the regularity data and a compatible pair of comparison/mass witnesses. It proves the full `QuantitativeRecurrence` specification. Its only substantial external predicates are SS, TV, KM-R and KM-Compatible. The same-parameter source selection includes the explicitly unformalized bridge KM-J. The generic finite construction's transition-provider premise is discharged internally, rather than assumed as an original conclusion. □

4 Pointwise comparison and unweighting

Definition D.4 (Orbit products and averages)**.** After choosing bounded representatives and scaling, suppose $|f_i| \le 1$. Define

$$z_n(x) = \prod_{i=1}^{\ell} f_i(T^{P_i(n-1)}x), \quad B_N^{\gamma} = N^{-1}\sum_{n=1}^{N}\Lambda_\gamma(n)z_n, \quad B_N = N^{-1}\sum_{n=1}^{N}\Lambda(n)z_n.$$

Then $|z_n| \le 1$.

Auxiliary lemma A.1 (Orbit transference from the finite estimate)**.** Assume TV. For each maximum degree $d \ge 1$, the same degree-only exponent $K \ge 1$ works as follows. Fix integer polynomials P_i of distinct positive increasing degrees, largest degree d. There is $C_{\mathbf{P}} > 0$, chosen before the system, its functions and N, such that every invertible probability system, essentially 1-bounded a.e. strongly measurable f_i, $N \ge 1$, and real weight g satisfying $|g(n)| \le B$, $B > 0$, and

$$\left|\sum_{n=1}^{N} g(n)\mathrm{e}(\phi(n))\right| \le NE \quad (\deg\phi \le d), \qquad E \ge 0,$$

satisfy integrability and the bound

$$\int_X \left|\frac{1}{N}\sum_{n=1}^{N} g(n)\prod_i f_i(T^{P_i(n-1)}x)\right| d\nu(x) \le C_{\mathbf{P}}B\big(N^{-1} + E/B\big)^{1/K}.$$

Status: checked derivation from TV of the required real-weight specialization of the related manuscript's Proposition 1.4; no independent own-paper premise.

Duality, exact spatial averaging and finite cutoffs. For exposition first suppose $|f_i| \le 1$ everywhere and put

$$A(x) = N^{-1}\sum_{n=1}^{N} g(n)\prod_i f_i(T^{P_i(n-1)}x), \qquad h(x) = \begin{cases}\overline{A(x)}/|A(x)|, & A(x) \ne 0,\\ 0, & A(x) = 0.\end{cases}$$

Then $|h| \le 1$ and $hA = |A|$. Let $L = N^d$ and $R_i(y) = P_i(y-1)$. Measure preservation gives

$$\int_X hA\,d\nu = \int_X \frac{1}{L}\sum_{a=1}^{L} h(T^a x)\frac{1}{N}\sum_{n=1}^{N} g(n)\prod_i f_i(T^{a+R_i(n)}x)\,d\nu(x).$$

For fixed x, use $F_0(a) = h(T^a x)\mathbf{1}_{[L]}(a)$ and restrict each $F_i(b) = f_i(T^b x)$ to a fixed multiple of $[-N^d, N^d]$, extending it by zero. A coefficient-sum bound chooses that multiple from $\mathbf{P}$ alone and contains every $a + R_i(n)$ used in the sum. The finite expression has exactly the normalization $L^{-1}N^{-1} = N^{-d-1}$. Apply the proved homogeneity form of TV and integrate the uniform bound on the probability space. For the original essentially bounded functions, the same pointwise estimate holds on the conull set where every integer iterate is bounded. □

The zero phase and PNT give

$$N^{-1}\sum_{n\le N}\Lambda_\gamma(n) = N^{-1}\sum_{n\le N}\Lambda(n) + O(N^{-\sigma_d}) \longrightarrow 1.$$

Proposition 4.1 (Weighted pointwise comparison; manuscript Proposition 4.1)**.** *After decreasing $c_* > 1$, for every $1 < c < c_*$ one has $B_N^{\gamma} - B_N \to 0$ almost everywhere.*

Status: checked conditional proof from SS, TV and PNT.

Power saving, subsequence and interpolation. Apply the orbit estimate with $g = D_\gamma$ and

$$B = C_c N^{1-\gamma} \log(2N).$$

Theorem 2.1 then gives

$$\|B_N^\gamma - B_N\|_1 \ll \log(2N) \left[N^{1-\gamma-1/K} + N^{(1-\gamma)(1-1/K)-\sigma_d/K} \right] \ll N^{-\eta}.$$

It suffices that $1-\gamma \le \min(1,\sigma_d)/(2K)$; one can take $\eta = \min(1,\sigma_d)/(4K)$ after absorbing the logarithm. Choose $A > \max(1, \eta^{-1})$, $N_j = \lfloor j^A \rfloor$. The L^1 norms are summable on this subsequence, so Tonelli gives a.e. convergence to zero there. For either nonnegative weight w, if $N_j \le N < N_{j+1}$,

$$|B_N^w - B_{N_j}^w| \le \frac{N_{j+1}-N_j}{N_j^2} \sum_{n \le N_j} w(n) + \frac{1}{N_j} \sum_{N_j < n \le N_{j+1}} w(n) = o(1).$$

The last step uses only the full mass asymptotic and $N_{j+1}/N_j \to 1$. Apply it to both weights. □

From weighted to prime-counting averages

By MW the ordinary weighted averages B_N converge a.e.; thus Proposition 4.1 gives a common limit $L(x)$ for B_N^γ. Higher prime powers satisfy

$$\sum_{\substack{p^k \le N \\ k \ge 2}} (\Lambda_\gamma(p^k) + \Lambda(p^k)) \ll_\gamma N^{3/2-\gamma} (\log(2N))^2 = o(N)$$

when $1/2 < \gamma \le 1$. The checked cardinality bound is $\sqrt{N}\,\log_2 N$, and the combined mass bound has coefficient $(\gamma^{-1}+1)/\log 2$. The actual normalized mass tends to zero in `PrimePowers.lean`. Removing it leaves primes exactly. Center at $L(x)$ and use $a(t) = \gamma^{-1} t^{1-\gamma} \log t$. Then $W(t) = \sum_{p \le t, p \in \mathbb{P}_c} a(p) \sim t$ and the centered weighted sum is $o_x(W(t))$. Summation by parts, $a' > 0$, and the same identity for W bound the unweighted centered sum by $C_{x,\varepsilon} + \varepsilon \pi_c(N)$. The mass asymptotic forces $\pi_c(N) \to \infty$, so division gives convergence to $L(x)$. The ordinary-prime calculation uses $a(t) = \log t$. The actual mass, prime-power residual, a.e. assembly and cutoff unweighting applications are checked.

Checked assembly of Theorem 1.2. `pointwiseErgodicTheorem_of_external_inputs` combines the preceding arithmetic, interpolation and unweighting steps and the bounded-representative transfer. Its only substantial external premises are SS, TV, PNT and MW. The orbit bound and its former OR-B translation are derived from TV in the preceding auxiliary lemma. Only the published finite-estimate bridge TV-B and the ordinary-convergence specialization MW-B remain external at this stage. □

5 Revision: lacunary variation of the weighted averages

The source supplied on 2 November 2026 adds Corollary 4.2, labelled `c:long-variation`. The averages remain the weighted averages B_N^γ and B_N of Section 4. No variational assertion for the count-normalized unweighted averages is claimed. The admissible c_* is selected before the variation exponent and the lacunarity ratio.

Definition D.5 (Infinite variation seminorm)**.** For $1 \le r < \infty$ and $a_j \in \mathbb{C}$, set

$$V^r(a) = \sup_{J \ge 1,\, j_0 < \cdots < j_J} \left(\sum_{h=1}^{J} |a_{j_h} - a_{j_{h-1}}|^r \right)^{1/r}.$$

The Lean definition takes this supremum in $[0,\infty]$, so infinite variation is not assigned an arbitrary real value. Sequence indices start at zero in Lean; shifting them by one gives exactly the manuscript's indexing. The extra $J = 0$ term is zero and does not change the supremum.

Status: checked definition and finite-chain correspondence with the real manuscript formula.

Auxiliary lemma A.2 (Summable perturbations and lacunary scales)**.** For $r \geq 1$, complex sequences satisfy

$$V^r(a+b) \leq V^r(a) + V^r(b), \qquad V^r(a) \leq 2\sum_{j\geq 0} |a_j|.$$

If $N_j \in \mathbb{N}_{>0}$ and $N_{j+1} \geq \lambda N_j$ with $\lambda > 1$, then for every $\eta > 0$,

$$\sum_{j\geq 0} N_j^{-\eta} \leq \sum_{j\geq 0} \lambda^{-j\eta} = \frac{1}{1-\lambda^{-\eta}}.$$

Consequently, for almost-everywhere measurable $E_j : X \to \mathbb{C}$,

$$\int_X V^r(E_j(x))\, d\nu(x) \leq 2\sum_{j\geq 0} \int_X |E_j(x)|\, d\nu(x).$$

Status: checked perturbation, measurability and infinite geometric-sum proofs.

Perturbation and geometric summation. Apply finite-dimensional Minkowski, bound each chain endpoint sum by the full nonnegative sum, and use Tonelli. Induction gives $N_j \geq \lambda^j$; comparison with the geometric series gives the uniform bound. □

External input E.1 (Ordinary-prime lacunary variation)**.** For each fixed distinct-degree integer polynomial family, $r > 2$ and $\lambda > 1$, there is a finite constant $C_{\mathbf{P},r,\lambda}$ such that, uniformly over probability systems, essentially 1-bounded inputs and positive λ-lacunary sequences,

$$\|V^r(B_{N_j} : j \geq 0)\|_1 \leq C_{\mathbf{P},r,\lambda}.$$

Status: external MW-V, not a project proof. Mirek–Wan–Wright, arXiv:2505.15549v2, Theorem 1.1(iv), equation (1.5), provides the published estimate. Its source-to-interface translation MW-V-B is recorded separately in the shared catalogue. Ordinary pointwise convergence alone does not supply this bound.

Theorem 5.1 (Lacunary variation; manuscript Corollary 4.2)**.** *For the fixed family* $\mathbf{P}$*, there is* $c_* > 1$ *such that, for every* $1 < c < c_*$*,* $r > 2$ *and* $\lambda > 1$*, every probability-preserving invertible system and* $f_i \in L^\infty(X)$ *satisfy*

$$\left\| V^r(B^\gamma_{N_j} : j \geq 0) \right\|_1 \leq C_{c,\mathbf{P},r,\lambda} \prod_{i=1}^{\ell} \|f_i\|_\infty$$

for every sequence of positive integers with $N_{j+1} \geq \lambda N_j$*. Here* $\gamma = 1/c$*, and the constant is chosen before the sequence, system and functions.*

Status: checked conditional proof from SS, TV and MW-V.

Transfer argument. For essentially 1-bounded inputs put $E_j = B^\gamma_{N_j} - B_{N_j}$. The already checked all-scale estimate `ps_orbit_discrepancy_L1_gamma` supplies $D > 0$ and $\eta > 0$ with $\|E_j\|_1 \leq DN_j^{-\eta}$. Hence

$$\|V^r(E_j)\|_1 \leq \frac{2D}{1-\lambda^{-\eta}}.$$

Apply MW-V to B_{N_j} and use the variation triangle inequality. For general inputs divide each nonzero input by its essential norm and use multilinearity. If one essential norm is zero, that input vanishes almost everywhere; measure preservation and countability give vanishing of every sampled product on a common conull set. Thus the zero-factor case also satisfies the stated product bound. □

External dependencies and manuscript correspondence

External input E.2 (KMTT comparison and weak regularity)**.** KM-U and KM-R are separate external finite estimates with the parameter ranges recorded in Proposition 3.1 and the shared catalogue. The compatible input also selects the prime-mass data with the same comparison parameters, including the external bridge KM-J. None assumes a PS-prime configuration or the conclusion of Theorem 1.1.

External input E.3 (Ordinary weighted pointwise convergence)**.** MW gives a.e. convergence of the ordinary von Mangoldt-weighted averages under the finite-L^q hypotheses; the probability-space bounded-input specialization is a bridge.

External input E.4 (Prime number theorem)**.** $N^{-1}\sum_{n\le N}\Lambda(n)\to 1$. A pinned-Mathlib theorem can replace this external entry only after its exact normalization and hypotheses are checked.

Shared manuscript correspondence

This table is identical in the two blueprints. A specification is a proposition-valued definition, not a theorem establishing that proposition. A quoted mathematical input is not a Lean axiom. The declaration catalogue and actual build evidence in the repository determine the checked status of auxiliary declarations.

Paper	Source label	Implementation boundary
Thm. 1.1	`t:szemeredi`	`quantitative_recurrence`: checked full specification from SS, TV, KM-R and KM-Compatible.
Thm. 1.2	`t:pointwise`	`pointwiseErgodicTheorem_of_external_inputs`: checked full uniform specification from SS, TV, PNT and MW.
Remark 1.3	unlabelled	Conjectural finite-L^p extension and endpoint background; not a theorem claimed here.
Thm. 1.4	`t:teravainen`	External finite generalized von Neumann estimate (TV); no project proof.
Thm. 2.1	`t:shao-sun`	External directly quoted polynomial-phase input SS; integer and small-scale translation included externally.
Lemma 2.2	`l:progression-phase`	Checked conditional theorem `progressionPhase_of_shaoSun`; no additional arithmetic input beyond SS specialization.
Prop. 2.3	`p:comparison`	`ps_counting_comparison`: exact original finite estimate checked conditionally on SS and TV; TV-B external.
Prop. 3.1	`p:KMTT`	`kmtt_density_increment_normalized`: checked from KM-U and KM-R; literal coefficient-one length bound.
Lemma 3.2	`l:PS-discrepancy`	`ps_prime_discrepancy`: checked from SS, TV and compatible same-parameter prime-mass data.
Prop. 4.1	`p:pointwise-comparison`	`ps_orbit_comparison`: exact original comparison checked from SS, TV and PNT.
Cor. 4.2	`c:long-variation`	`lacunary_variation_of_external_inputs`: checked conditional deduction from SS, TV and MW-V; exact infinite variation and essential-norm product.

The revised source has ten labelled theorem-like statements and one unlabelled numbered remark. There are 49 labels in total: ten statement labels and 39 equation labels. The numbering is taken from the actual compiled source, including its global equation numbers (1)–(39). The seven theorem declarations in the table establish the original numbered deductions under their explicit premises; the two quoted inputs remain external.

Shared external-input catalogue

Eight explicit external predicates are implemented and uninstantiated: SS, TV, PNT, MW, MW-V, the standalone KM-U interface, KM-R and KM-Compatible. OR is a derived specification proved from TV, not an independent assumption about the author's other paper. Their actual names and types are in `ExternalInputs.lean`, `OrbitInputs.lean`, `KMTTInterfaces.lean` , `PrimeMassInputs.lean` and `LongVariation.lean`. The compatible predicate includes the same-parameter comparison and mass witnesses; the standalone KM-U predicate is not an extra premise when that compatible input is used. A catalogue entry is not a Lean axiom.

SS is the directly published polynomial-phase consequence of Shao–Sun, arXiv:2605.18676v1, Theorem 2.1 and Definition 4.5. **TV** is Teräväinen, arXiv:2401.03174v2, Theorem 4.1. The universal-phase/supremum bridge TV-B remains external. SS and TV give Proposition 2.3. TV also proves the orbit transference used by the pointwise branch.

KM-U is KMTT, arXiv:2608.19525v1, Proposition 5.2; **KM-R** is Lemma 5.3, with its small-scale alternative and coordinate-dependent resolutions. **KM-M** records (5.2) and (5.15). The actual `KMTTCompatibleInput` selects the same comparison and mass parameters jointly; **KM-J** is its explicitly unformalized fixed-cutoff and constant-reselection bridge. The standalone `KMTTUnweightedInput` is subsumed when this joint input is used. Only recurrence uses these predicates.

OR is derived from TV: the required real-weight specialization of the related sequences-paper Proposition 1.4 is proved internally, including shifts, support, essential bounds and integrability. **MW** is Mirek–Wan–Wright, arXiv:2505.15549v2, Theorem 1.1(ii), via `OrdinaryPrimePointwiseInput`; its source translation MW-B remains external. **PNT** is the ordinary von Mangoldt mass limit, directly stated in Tao's 246B Notes 4, Theorem 6. Theorem 1.2 has exactly SS, TV, PNT and MW as its substantial external premises. The new MW-V interface quotes MWW Theorem 1.1(iv), equation (1.5); its finite-L^q, Hölder and full-norm-to-seminorm translation is external MW-V-B. Corollary 4.2 uses exactly SS, TV and MW-V, without PNT. The original main-theorem premises are unchanged. The optional no-Siegel footnote is not used.

Primary sources were read in the cited versions. Provenance verification does not formalize an external theorem or discharge its source-formulation bridges. See the repository assumption catalogue and reference review for the actual interfaces.

Verified scope and limitations

All seven original numbered deductions and new Corollary 4.2 are checked conditionally; Theorems 1.4 and 2.1 remain external. The eight explicit external predicates are uninstantiated. OR is derived from TV. The remaining source-formulation bridges, including joint KMTT parameter selection KM-J, remain external. Full unconditional formalization of the quoted analytic ingredients is outside the verified scope.